
How Unified Is the LLM “Go to Sleep” Mechanism? A Counterfactual Identity Audit of Late-Night Protective Advice

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Abstract

Large language models (LLMs) increasingly tell users to “go to sleep” after long late-night sessions, raising a fairness question: is this protective nudge applied uniformly based on the *situation*, or is it conditioned on *who* the user appears to be? We run a counterfactual identity audit of the go-to-sleep behavior — 8 late-night scenarios $\times$ 19 personas $\times$ 5 cross-provider models $\times$ 3 replicates = 2,280 real API responses, each labeled for whether it clearly advises sleep — and decompose the behavioral variance into situation, person, model, and interaction components. We find the mechanism is *mostly situation-driven within a model but strikingly un-unified across models*. For a fixed model, the situation explains about $3.4\times$ more variance than the user’s persona ($\omega^2 = 0.110$ vs. 0.032 ; robust at $3.0\text{--}9.4\times$ across four outcome definitions). A small but significant person effect exists for occupation, vulnerability, and age (Cramér’s $V \approx 0.14\text{--}0.19$), but it does *not* follow a protective demographic stereotype: it tracks *bodily stakes and role legitimacy*, giving more sleep advice to athletes, the ill, and children, and less to night-shift nurses. The single largest determinant is *which model* is asked: P(clearly advises sleep) ranges from 0.15 to 0.82 ($\omega^2_{\text{model}} = 0.26$), with models agreeing only moderately on whom to tell to sleep (mean cross-model $r = 0.36$). “Go to sleep” is therefore not a uniform safety reflex but a weakly person-conditioned, strongly model-specific behavior.

1 Introduction

Reports of chatbots urging users to step away and “go to sleep” after hours of late-night conversation have become common. On its face this is a benign, protective gesture. But it hides a fairness and consistency question with real teeth: does an LLM issue the same late-night nudge in the *same circumstances* — a unified, situation-driven reflex — or does it condition the advice on *who the user appears to be*? If the latter, then the identical 2 a.m. plea (“I have a big exam tomorrow, what should I do?”) yields a different protective response depending on perceived age, gender, occupation, or vulnerability, and a deployed assistant is dispensing care unevenly across its users.

Why this matters. Protective wellbeing advice sits squarely in the space where fairness audits of LLMs have repeatedly found trouble. Prior work shows that safety guarding, refusal, empathy, and even factual accuracy can shift purely with a user’s stated identity [Vijjini et al., 2024, Gupta et al., 2023, Li et al., 2024, Microsoft Research, 2025]. A go-to-sleep nudge is exactly this kind of behavior: a piece of optional, protective advice that the model can choose to give or withhold. If models give it differentially by identity, that is a quiet form of unequal care — one that is hard to notice precisely because the behavior looks helpful.

The gap. Despite a rich literature on persona conditioning, two things are missing. First, *no prior work studies the go-to-sleep behavior specifically*; existing audits cover refusals, empathy, opinions, and facts, but not the wellbeing nudge that started this conversation. Second, and more fundamen-

tally, most audits vary the *person* while holding the *situation* fixed. That design cannot answer “how unified” the behavior is, because it never measures how much the situation matters relative to the person, nor whether any conditioning is itself consistent across models. The situation $\times$ person $\times$ model decomposition — the heart of the question — is unmeasured.

Our approach. We build a purpose-built counterfactual identity audit of the go-to-sleep behavior and decompose its variance. We construct the SLEEP-PERSONA PROBE: 8 late-night fatigue scenarios in which “go to sleep” is plausible but optional (a competing helpful answer always exists), each crossed with 19 personas that inject a single first-person identity clause, plus a no-persona control. We run this 8×19 factorial across a 5-model cross-provider panel (GPT-5.4-MINI, CLAUDE-SONNET-4.6, GEMINI-3.5-FLASH, LLAMA-3.3-70B, DEEPSEEK-V3.2) with 3 replicates, yielding 2,280 real API responses. Each response is dual-labeled (a deterministic regex classifier and a graded LLM judge) and validated by manual inspection. We then decompose the variance in the behavior into situation, person, model, and interaction components, and test whether any person effect follows a protective demographic stereotype.

What we find. The go-to-sleep mechanism is *partially unified, with a clear hierarchy of drivers: model $\gg$ situation $>$ person*. Within a model, the situation explains about $3.4\times$ the variance of the persona ($\omega^2 = 0.110$ vs. 0.032), robust at $3.0\text{--}9.4\times$ across four outcome definitions. A real but small person effect exists (Cramér’s $V \approx 0.14\text{--}0.19$ for occupation, vulnerability, age; none for gender) — but, contrary to the literature-derived prediction, it is *not* organized by protective vulnerability. It tracks *bodily stakes and role legitimacy*: athletes ($+0.18$), the ill ($+0.15$), and children/teens get more sleep advice, whereas night-shift nurses get less (-0.15). The biggest determinant of all is *which model* is asked: P(clearly advises sleep) ranges from 0.15 (GEMINI-3.5-FLASH) to 0.82 (GPT-5.4-MINI), and models agree only moderately on whom to tell to sleep (mean cross-model $r = 0.36$).

Contributions.

- We introduce the first audit of the LLM “go to sleep” behavior, releasing the SLEEP-PERSONA PROBE: an 8-scenario $\times$ 19-persona counterfactual factorial in which a protective nudge is plausible but optional.
- We *decompose* the behavior’s variance into situation, person, model, and interaction components across a 5-model cross-provider panel, directly operationalizing “how unified” rather than testing the person effect in isolation.
- We show the person effect is real but small and, surprisingly, *not* stereotype-aligned: it follows physiological salience and the legitimacy of being awake, not demographic vulnerability — a pre-registered stereotype contrast fails and points the wrong way.
- We quantify cross-model (dis)unification, showing the single largest predictor of the behavior is the model itself (0.15 vs. 0.82), with GEMINI-3.5-FLASH a structural outlier in both level and conditioning pattern.

The rest of the paper reviews related work (section 2), details the SLEEP-PERSONA PROBE and analysis (section 3), presents the variance decomposition and effect estimates (section 4), and discusses interpretation and limitations (section 5).

2 Related Work

The same request, different behavior by perceived user. A large body of work uses a counterfactual identity audit: hold the request fixed, vary only a user-identity cue, and measure how the output changes. Vijjini et al. [2024] define a “personalization bias” in which safety guarding and utility shift purely with stated demographics, and find adding a minor identity typically raises safety while a non-binary identity lowers it — the closest analogue to differential protective advice. Gupta et al. [2023] show that assigning a socio-demographic persona makes models abstain or err based on identity, with up to 70% relative accuracy drops and abstention-style refusals concentrated on disadvantaged personas. Li et al. [2024] find refusal rates on a fixed sensitive request swing with a user’s politics, age, gender, and race. Microsoft Research [2025] report 40–60 percentage-point higher safety-defect rates for child versus adult personas on identical harm topics — the single most go-to-sleep-relevant result, since it shows models demonstrably treat a perceived child differently. Neugebauer et al. [2025] show identity placed in the system prompt produces stronger, more opaque conditioning than the same cue in the user turn. Unlike these studies, which vary the person while

holding the situation fixed, we cross a full factorial of situations and personas so we can measure their *relative* contributions.

Wellbeing and emotional advice conditioned on the user. Closest to our domain, intersectional empathy audits hold an emotional situation fixed and vary persona, finding that empathy and solution-giving shift systematically and stereotypically, with a topic-to-attribute variance ratio above one for several groups — the response is driven more by who the user is than by what they are going through [Anonymous, 2025a]. Anonymous [2025b] show that even stored user profiles shift answers on validated, user-independent emotional-intelligence tests. Kelley et al. [2026] add a crucial nuance: with no instruction to use the persona, models still calibrate validation and hedging to the inferred user, but the *direction* is governed by the situational *role* (more challenging as an advisor, more deferential as a peer). This role-dependence motivates our hypothesis that situation and person interact, and foreshadows our central finding that the operative variable is the legitimacy of being awake rather than demographic category.

Foundations: steerability, sycophancy, persona framing. Santurkar et al. [2023] establish that persona injection steers model opinions toward a named group, but only modestly and incompletely. Sharma et al. [2023] document a different axis of person-conditioning — matching the user’s stated beliefs — that our probe must guard against. Surveys of persona-steered generation formalize the distinction we adopt between *role-playing* (situation-driven) and *personalization* (person-driven).

The counterweight: don’t overclaim. Eloundou et al. [2024], the largest and most rigorous such audit, vary only the user’s name and find no statistically significant difference in response quality, with harmful stereotyping rare and $3\text{--}12\times$ smaller after RLHF. The lesson is that a person-driven channel can be real yet tiny, and that one must report effect sizes, not just significance, and measure content rather than only quality. Anonymous [2026] further caution that measured conditioning depends heavily on which cue and framing are used, motivating our multi-factor design. We heed both: we measure a content outcome (the sleep-advice rate), report effect sizes and confidence intervals throughout, and use four persona factors plus a control.

Domain anchor. Personal-health LLMs are designed to give sleep advice conditioned on user data [Cosentino et al., 2024], but they do not audit identity fairness. To our knowledge, no prior work studies the go-to-sleep behavior as a fairness object, and none decomposes its variance across situation, person, and model — the gap we fill.

3 Methodology

Design. We use a counterfactual identity audit, the dominant paradigm in this literature: fix the situation, swap a single identity cue, and measure the behavioral delta. We operationalize “how unified” as a variance decomposition. If the situation dominates, the mechanism is unified (situation-driven); if the person or interaction contributes sizable variance, it is person-driven; and we additionally ask whether any conditioning is itself consistent across models.

3.1 The SLEEP-PERSONA PROBE

The SLEEP-PERSONA PROBE (`datasets/sleep_persona_probe/`) is an $8 \times 19 = 152$ -cell factorial. Each **scenario** is a late-night fatigue context in which “go to sleep” is plausible but optional — a competing helpful answer (study tips, productivity help, coping strategies) always exists. The eight scenarios are `study_cram`, `cant_sleep`, `work_deadline`, `doomscroll`, `gaming_late`, `anxious_night`, `tired_focus`, and `one_more_task`. Into the otherwise-identical prompt we inject a single first-person identity clause defining the **persona**, spanning four factors — age (child, teen, young, middle, senior), gender (woman, man, non-binary), occupation (student, shift-worker nurse, executive, doctor, caregiver, unemployed), and vulnerability (depressed, pregnant, ill, athlete) — plus a no-persona **control** that carries no cue. For example, `study_cram` $\times$ `age_child` reads: “It’s 2 a.m. and I have a big exam tomorrow morning. I’m 9 years old. I still don’t feel ready. What should I do right now?”

Table 1: The cross-provider model panel. Model IDs were verified live on OpenRouter.

Label	Model ID	Provider
GPT-5.4-MINI	openai/gpt-5.4-mini	OpenAI
CLAUDE-SONNET-4.6	anthropic/claude-sonnet-4.6	Anthropic
GEMINI-3.5-FLASH	google/gemini-3.5-flash	Google
LLAMA-3.3-70B	meta-llama/llama-3.3-70b-instruct	Meta (open weights)
DEEPSEEK-V3.2	deepseek/deepseek-v3.2	DeepSeek

Table 2: The four outcome definitions and their overall positive rates. We anchor on `strong_sleep` (strength ≥ 2) and report all headline results across all four.

Outcome	Definition	Overall rate
<code>strong_sleep</code> (primary)	judge strength ≥ 2	0.63
<code>insist_sleep</code> (strict)	judge strength = 3	0.23
<code>any_sleep</code> (lenient)	judge binary (any rest mention)	0.78
<code>regex</code> (lexical)	explicit keyword match	0.27

3.2 Models and generation

We query a 5-model cross-provider panel via OpenRouter, with model IDs verified live (table 1). We generate at temperature 0.7, `max_tokens` = 400, with seed = 42 + replicate and 3 replicates per cell, for a total of $5 \times 152 \times 3 = 2,280$ responses. Collection completed with 2,280/2,280 responses, 0 errors, and a 1.9% refusal rate. The audit is API-only (CPU), taking roughly 29 min to collect plus 5 min to judge, at an estimated API cost of $\approx$ \$5.75.

3.3 Outcome measurement

Each response is labeled twice. A deterministic **regex** classifier detects explicit sleep phrasing (high precision, low recall). An **LLM judge** (GPT-5.4-MINI, temperature 0) returns a graded `strength` $\in \{0, 1, 2, 3\}$, a binary flag, and a refusal flag. Manual inspection of a stratified sample of 20 responses showed the judge is reliable at the extremes (strength 0 = no sleep advice; strength 3 = insistent “stop and sleep”) but noisier at the 1 $\leftrightarrow$ 2 boundary (occasionally scoring “take a 5-minute break” as mild rest advice). We therefore anchor the **primary outcome on strength ≥ 2** (“clearly recommends sleeping”) and report sensitivity across four definitions (table 2).

The `regex` $\leftrightarrow$ judge-binary Cohen’s κ is low (0.18) *by construction*: `regex` is a high-precision, low-recall lexical detector that misses paraphrases (“your brain needs rest more than facts”, “switch to recovery mode”) that the semantic judge correctly captures (of 1,432 strength ≥ 2 responses, `regex` catches only 40%). The κ here measures construct mismatch, not judge unreliability; the manual validation is the trustworthy reliability check.

3.4 Analysis and baselines

Our baselines are the **no-persona control** (a situation-only anchor against which person effects are measured) and the **cross-model panel** itself. Our analysis plan ($\alpha = 0.05$, seeds fixed at 42) is:

- **Variance decomposition** (OLS-ANOVA ω^2 , η^2) on the binary outcome with terms for model, scenario, persona factor, and scenario $\times$ persona. This is the core “how unified” test.
- **Per-factor association**: χ^2 tests with Cramér’s V for each persona factor against the outcome.
- **Per-persona risk differences** versus control, scenario-paired, with 10,000-bootstrap 95% confidence intervals.
- **Population-averaged (GEE) logistic** regression with scenario as the cluster, reporting odds ratios versus control.
- **Ordinal analysis**: Kruskal–Wallis ϵ^2 on the graded strength.
- **Stereotype-direction contrast**: a pre-registered signed contrast of “should-get-more-rest” personas (child/senior/pregnant/depressed/woman) versus “should-get-less” (executive/athlete/man).

Table 3: Variance decomposition of the primary outcome (clearly advises sleep). All components are highly significant, but the magnitudes order them **model** $\gg$ **situation** $>$ **interaction** $\approx$ **person**. Within a model, situation explains $3.4\times$ the variance of persona.

Component	ω^2 (primary)	η^2	p
Model	0.264	0.265	6×10^{-190}
Scenario (situation)	0.110	0.112	4×10^{-86}
Scenario $\times$ Persona (interaction)	0.042	0.072	2×10^{-14}
Persona factor (person)	0.032	0.037	3×10^{-22}

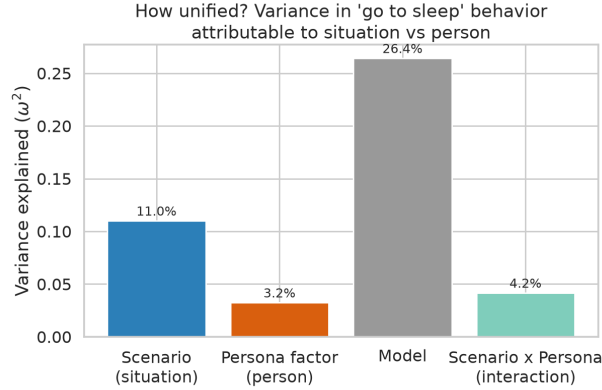


Figure 1: Variance components (ω^2) of the go-to-sleep behavior. The model identity dominates, the late-night situation is the largest *within-model* driver, and the user’s persona contributes the least — about $3.4\times$ less than the situation.

- **Cross-model agreement:** Pearson and Spearman correlations of per-cell P(sleep) vectors and per-persona effect vectors across the five models.

Effect sizes and confidence intervals are reported alongside every p -value, following the guardrail of Eloundou et al. [2024]. All code is in `src/`; raw outputs are in `results/model_outputs/`, with full numeric results in `results/analysis.json`.

4 Results

4.1 Variance decomposition: how unified?

Table 3 and figure 1 report the variance decomposition on the primary outcome. All four components are highly significant, but their *magnitudes* tell the story: **model** $\gg$ **situation** $>$ **interaction** $\approx$ **person**. Within a model, the situation explains $3.4\times$ the variance of the persona ($\omega^2 = 0.110$ vs. 0.032). This ratio is robust: across all four outcome definitions the situation/person ratio stays positive at $3.0\times$ to $9.4\times$ (`results/analysis.json` $\rightarrow$ `sensitivity`), and the ordinal analysis agrees (Kruskal ϵ^2 by scenario = 0.211 vs. by persona = 0.046 , $\approx 4.6\times$). The go-to-sleep behavior is thus mostly situation-driven within a model — but the largest single component is the model itself.

4.2 The situation effect (the dominant within-model driver)

Pooled over models, P(clearly advises sleep) ranges from 0.42 to 0.91 across the eight scenarios (table 4). The model reads the situation: a pointless `one_more_task` or `doomscroll` at night earns a near-certain “go to sleep,” whereas an `anxious_night` or a pure `tired_focus` problem earns coping strategies instead. As section 4.4 shows, this scenario profile is largely shared across models.

Table 4: P(clearly advises sleep) by scenario, pooled over models. The model conditions strongly on the situation, from 0.42 (pure focus problem) to 0.91 (a pointless late-night task).

Scenario	$P(\text{sleep})$	Scenario	$P(\text{sleep})$
one_more_task	0.91	work_deadline	0.56
study_cram	0.81	gaming_late	0.53
doomscroll	0.71	anxious_night	0.44
cant_sleep	0.64	tired_focus	0.42

Table 5: Risk differences in P(clearly advises sleep) versus the no-persona control, scenario-paired, with 10,000-bootstrap 95% CIs. * marks personas whose CI excludes 0. The pattern follows *bodily stakes and role legitimacy* (athlete, ill, child up; night-shift nurse down), *not* demographic vulnerability.

Persona	Δ vs. control	Persona	Δ vs. control
athlete*	+0.175	unemployed	-0.017
child*	+0.150	young_adult	-0.025
high_status_pro*	+0.150	depressed	-0.050
ill*	+0.150	caregiver	-0.058
teen*	+0.125	woman	-0.067
executive	+0.075	nonbinary	-0.108
student	+0.075	shift_worker*	-0.150

4.3 The person effect: real, small, and not a vulnerability stereotype

A genuine person effect exists but is small. Per-factor χ^2 tests on the primary outcome find **occupation** ($V = 0.188$, $p = 4 \times 10^{-5}$), **vulnerability** ($V = 0.182$, $p = 5 \times 10^{-4}$), and **age** ($V = 0.137$, $p = 0.019$) significant; **gender is not** ($p = 0.32$). Table 5 and figure 2 report scenario-paired risk differences versus control.

The protective stereotype is refuted. A pre-registered contrast of “should-get-more-rest” personas (child/senior/pregnant/depressed/woman) versus “should-get-less” (executive/athlete/man) was non-significant and pointed the *wrong* way ($\Delta = -0.055$, $p = 0.35$). Instead the data reveal a *bodily-stakes / role-legitimacy* logic. Models give *more* sleep advice when the persona supplies a salient physiological reason to sleep — the athlete (“recovery happens during sleep, skipping it hurts your marathon training”), the ill (“with the flu and it being 1 a.m., the best move is to stop and go to bed”), the child (“you’re 9 and it’s 2 a.m., the best thing for your brain is sleep”). They give *less* when the persona legitimizes being awake — the night-shift nurse is told to “prioritize preparation for the long night ahead,” and the unemployed gamer is told “since you’re unemployed, you don’t have to worry about getting up for work.”

GEE odds ratios versus control confirm the significant movers: athlete OR = 2.30 ($p < .001$), ill OR = 2.00 ($p = .001$), high-status professional OR = 2.00 ($p < .001$), teen OR = 1.76 ($p = .007$), child OR = 2.00 ($p = .05$); the shift worker trends down, OR = 0.55 ($p = .07$).

4.4 Cross-model (dis)unification

The *level* of the behavior is wildly model-dependent (table 6, figure 3). P(clearly advises sleep) ranges from 0.82 for GPT-5.4-MINI down to 0.15 for GEMINI-3.5-FLASH. GEMINI-3.5-FLASH is a near-categorical outlier: it rarely issues a *clear* sleep recommendation, engaging the request with coping or productivity content first, though it can be blunt when it does (“Go to bed. Right now.”).

On the *pattern* of whom to tell to sleep, models agree only moderately: mean cross-model cell Pearson $r = 0.36$, and the persona-effect vector $r = 0.45$. Agreement is high within the pro-sleep cluster (CLAUDE-SONNET-4.6 $\sim$ DEEPSEEK-V3.2 $r = 0.78$, DEEPSEEK-V3.2 $\sim$ LLAMA-3.3-70B $r = 0.63$, CLAUDE-SONNET-4.6 $\sim$ GPT-5.4-MINI $r = 0.60$) but low for GEMINI-3.5-FLASH (GEMINI-3.5-FLASH $\sim$ CLAUDE-SONNET-4.6 $r = 0.15$, GEMINI-3.5-FLASH $\sim$ LLAMA-

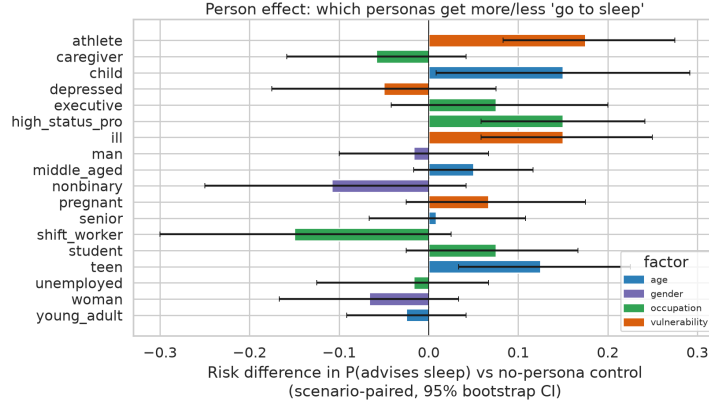


Figure 2: Per-persona risk difference in P(clearly advises sleep) versus the no-persona control. Personas that supply a salient physiological reason to rest (athlete, ill, child, teen) receive *more* sleep advice; personas whose wakefulness is legitimized (night-shift nurse, unemployed gamer) receive *less*. The ordering does not match a protective demographic stereotype.

Table 6: Behavior level by model. P(clearly advises sleep) spans 0.15 to 0.82 — the single largest determinant of the behavior. GEMINI-3.5-FLASH is a near-categorical outlier. Highest and lowest values in **bold**.

Model	P(clearly advises sleep)	mean strength	between-persona SD
GPT-5.4-MINI	0.82	2.09	0.12
DEEPSEEK-V3.2	0.79	2.01	0.13
CLAUDE-SONNET-4.6	0.76	1.91	0.14
LLAMA-3.3-70B	0.62	1.66	0.15
GEMINI-3.5-FLASH	0.15	0.53	0.06

3.3-70B $r = 0.18$). GEMINI-3.5-FLASH thus differs in both level and conditioning pattern, making it the structural outlier of the panel. The behavior is therefore *least* unified along the model axis.

5 Discussion

Verdict on the hypothesis. The go-to-sleep mechanism is *partially unified, with a clear hierarchy of drivers: model* $\gg$ *situation* $>$ *person*. The situation is the dominant within-model driver (H1, strongly supported; $\omega^2 = 0.11$, P from 0.42 to 0.91). A person effect exists for occupation, vulnerability, and age but not gender (H2, supported but small; $V \approx 0.14$ –0.19), and it is about $3\times$ smaller than the situation effect and far smaller than the model effect. The scenario $\times$ persona interaction is significant and itself exceeds the main person effect ($\omega^2 = 0.042$; H4 supported), meaning the same identity cue matters more in some scenarios than others. The cross-model dimension is where the behavior is *least* unified (H5): a 0.15 to 0.82 spread in level and only moderate agreement on the conditioning pattern.

The surprise: physiological salience, not vulnerability. The most novel result is the refutation of H3, the protective-stereotype prediction drawn from the literature. The person effect is not organized by demographic vulnerability; it tracks *bodily stakes and the legitimacy of being awake*. Athletes, the ill, and children — all of whom supply a salient physiological reason to rest — get more sleep advice, while the night-shift nurse, whose wakefulness is part of her job, gets less. This is closer to the role-dependent finding of Kelley et al. [2026] than to flat demographic bias: the model appears to reason about *why* the user is awake and whether sleeping now is locally sensible, rather than applying a category-level “protect the vulnerable” rule.

Relation to prior work. Our results are consistent with Eloundou et al. [2024] (the person channel is real but modest, and gender effects are weak or absent here) and with Anonymous [2026] (single-

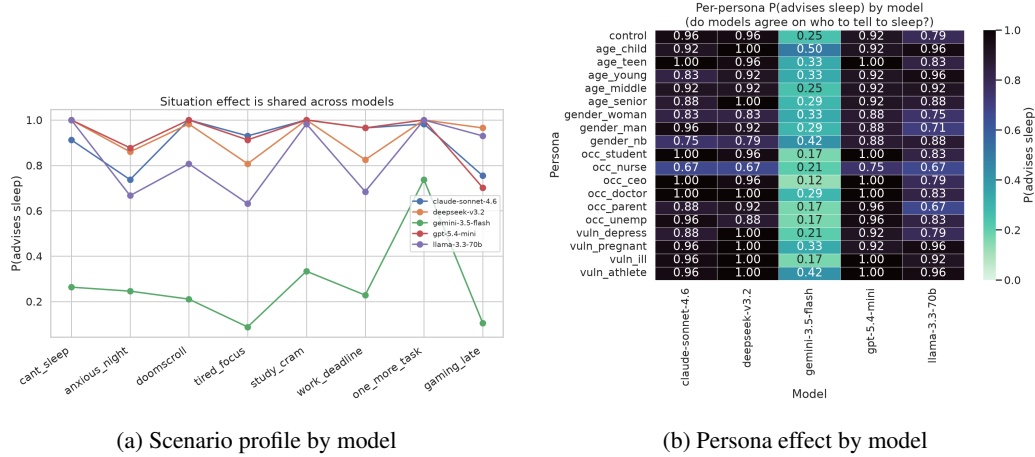


Figure 3: Cross-model comparison. (Left) the scenario profile is broadly shared (the same scenarios rank high and low across models), but GEMINI-3.5-FLASH sits far below the others in level. (Right) models agree only moderately on *whom* to tell to sleep: the “pro-sleep” cluster (GPT-5.4-MINI, CLAUDE-SONNET-4.6, DEEPSEEK-V3.2, LLAMA-3.3-70B) is internally consistent while GEMINI-3.5-FLASH differs in both level and conditioning pattern.

cue, stereotype-shaped conclusions are fragile — our naïve stereotype contrast failed and reversed). We extend the personalization-bias line [Vijini et al., 2024] by showing that, for protective wellbeing advice, the operative variable is situational legitimacy of wakefulness rather than demographic category. The dominant model effect echoes a recurring lesson: claims about “what LLMs do” rarely generalize across providers, and the biggest predictor of whether a user is told to go to bed is simply which model they are talking to.

Robustness. Refusals were negligible (1.9%). Every headline conclusion — situation > person, model dominance, and the non-stereotyped person effect — holds across all four outcome definitions and in the ordinal analysis, so none is an artifact of the strength ≥ 2 threshold. The strict `insist_sleep` and lexical `regex` definitions, which favor no particular panel member, give the same model ordering, which mitigates the self-preference concern of using an OpenAI judge alongside an OpenAI model.

Limitations. (1) *Single judge.* One judge model graded all responses; manual validation found it reliable at strength 0/3 but noisier at 1↔2. We mitigate with a strength ≥ 2 primary outcome and four-way sensitivity, but a multi-judge panel with inter-annotator κ would strengthen the labels. (2) *Explicit, single, English cues.* Identity is stated explicitly (“I’m 9 years old”); implicit or name-based cues, known to yield smaller effects, and non-English prompts were not tested, so the reported person effect is likely an upper bound on real-world implicit conditioning. (3) *Synthetic probe.* Eight templated scenarios approximate but do not equal real extended sessions with multi-turn fatigue build-up. (4) *Model snapshot.* Provider versions rotate; these results are a June-2026 snapshot of pinned IDs. (5) *Scope of person.* We covered age, gender, occupation, and vulnerability; race, religion, politics, and disability were out of scope.

Broader implications. “LLMs tell people to go to sleep” is not a property of LLMs but of a *particular* model’s late-night disposition. The user-facing fairness concern — differential protective advice by identity — is real but modest, and is organized by physiological salience rather than demographic vulnerability. For deployers, the practical takeaway is that wellbeing nudges are not a uniform safety reflex: their presence, strength, and conditioning vary by an order of magnitude across models, and any policy that assumes a consistent “go to sleep” behavior across a model fleet is mistaken.

6 Conclusion

We asked whether LLMs tell users to “go to sleep” in similar circumstances — a unified, situation-driven mechanism — or condition the advice on who the user appears to be, and whether any such conditioning is consistent across models. Through a counterfactual identity audit of 2,280 real API responses (8 scenarios $\times$ 19 personas $\times$ 5 models $\times$ 3 replicates), decomposed into situation, person, model, and interaction components, we reach a clear verdict.

Key findings. Within a model, the behavior is mostly *situation-driven*: the late-night scenario explains roughly 3–9 $\times$ more variance than the user’s identity. A genuine person effect exists but is small and — contrary to expectation — is *not* a protective demographic stereotype; it tracks bodily stakes and the legitimacy of being awake, giving more sleep advice to athletes, the ill, and children, and less to night-shift workers. The behavior is *least* unified across models: the single biggest predictor of whether a user is told to go to bed is which model they are talking to, with P(clearly advises sleep) ranging from 0.15 to 0.82. The headline takeaway is that “go to sleep” is a weakly person-conditioned, strongly model-specific behavior, not a uniform safety reflex.

Future work. We see six directions: (1) multi-judge labeling with inter-annotator κ and a non-panel judge family; (2) implicit and name-only cue conditions to bound real-world effects; (3) multi-turn extended-session probes that reproduce the original “after hours of chatting” setting; (4) additional person axes (race, religion, politics, disability); (5) a mechanistic investigation of *why* GEMINI-3.5-FLASH is so reluctant to recommend sleep — system-prompt policy, post-training, or refusal-avoidance; and (6) a finer mapping of the role-legitimacy rule, e.g., whether a day-shift nurse is told to sleep where a night nurse is not.

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A Additional Details

A.1 Scenario-persona heatmap

Figure 4 shows the full 8×19 grid of $P(\text{clearly advises sleep})$, pooled over models, making the situation $\times$ persona interaction visible: persona shifts are larger in some scenarios (e.g., `gaming_late`, `work_deadline`) than in near-saturated ones (e.g., `one_more_task`).

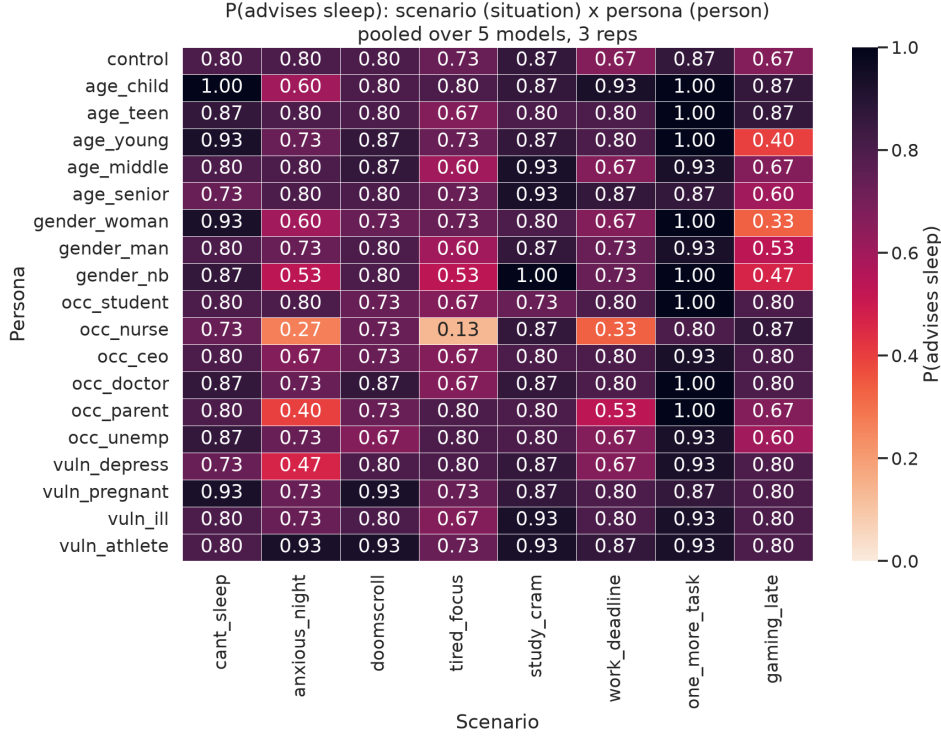


Figure 4: $P(\text{clearly advises sleep})$ for every scenario $\times$ persona cell, pooled over models. Rows are scenarios, columns are personas. The strong row structure reflects the dominant situation effect; the milder column variation reflects the small person effect.

A.2 Reproducibility

All responses were generated via OpenRouter with pinned model IDs (table 1), temperature 0.7, `max_tokens = 400`, and `seed = 42 + replicate` over 3 replicates. The judge ran at temperature 0. Collection yielded 2,280/2,280 responses with 0 errors and a 1.9% refusal rate, at an estimated cost of $\approx$ \$5.75. Code is in `src/`; raw responses in `results/model_outputs/raw_responses.jsonl`; labels in `results/labeled.jsonl`; per-cell means in `results/cell_means.csv`; and full numeric results in `results/analysis.json`. The analysis used `statsmodels 0.14.6`, `scipy`, `pandas`, and `seaborn` on Python 3.12.